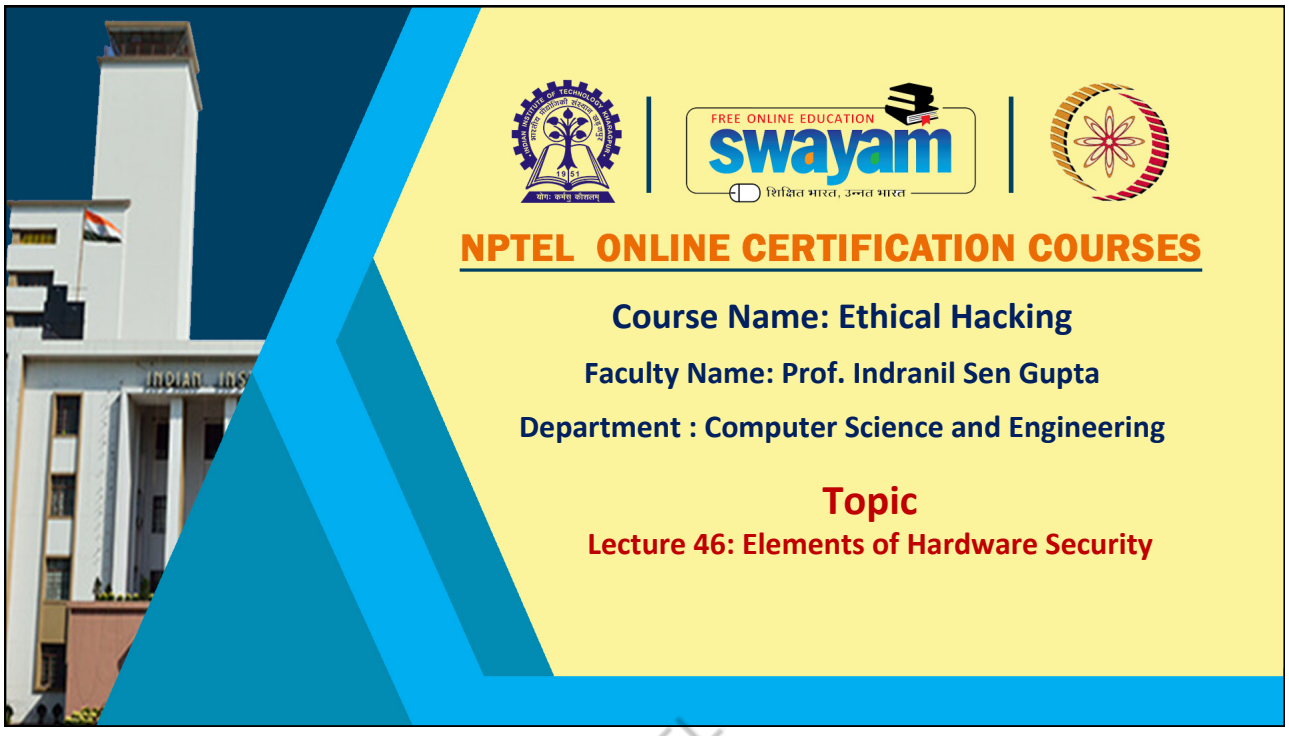


NPTEL ONLINE CERTIFICATION COURSES

Course Name: Ethical Hacking

Faculty Name: Prof. Indranil Sen Gupta

Department : Computer Science and Engineering

Topic

Lecture 46: Elements of Hardware Security

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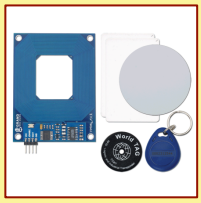
CONCEPTS COVERED

- Attacks on hardware
- Typical countermeasures

Introduction

- How to we characterize hardware in the present context?
  - **Computer Hardware**, which includes processors, firmware and memory.
  - **Mobile Hardware**:
    - ❖ SIM Card
    - ❖ RFID
    - ❖ Smart Card
    - ❖ PUF



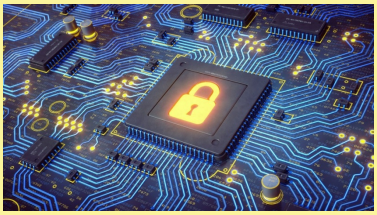


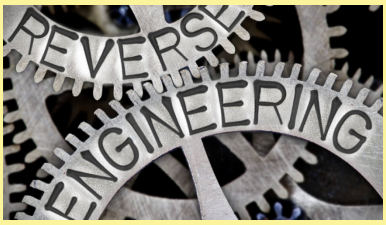


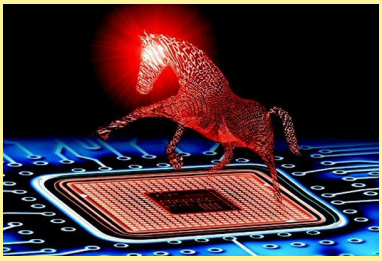
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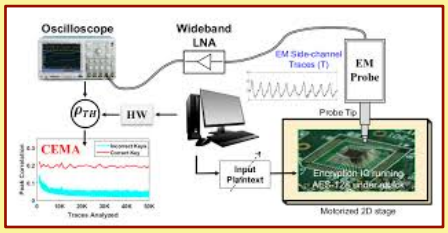
Attacks on Hardware

- **Physical Attacks**
  - Carried out on the actual device using hardware tools.
- **Planned Attacks**
  - Some vulnerability can be deliberately included in the hardware.
- **Stealing Secret Data**
  - Many hardware device carries confidential data.
  - International mobile subscriber identity and contact details in a SIM card.
  - Unique identification codes in RFID tags.
  - Secret key and other confidential information in a Smart Card.









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Types of Attacks

a) Black Box Testing

- The attacker sends an input to the circuit and receives an output.
- Based on the input/output behavior, the attacker decides what kind of algorithm is used.
- This is an non-invasive type of attack.

b) Physical Probing

- The attacker plants a probe into the chip itself and reads data off the chip.
- This is an invasive attack, and requires sophisticated instrumentation.

Types of Attacks

- **Reverse Engineering**
  - The attacker acquires the device (say, smart card) and physically exposes the circuit.
  - Each layer of the circuit is removed and high resolution photographs are taken.
  - Invasive approach, and also requires very sophisticated instrumentation.
- **Side Channel Analysis**
  - The attacker measures sensitive parameters during normal operation of the circuit.
  - Based on the measurements, some secret values can be inferred.
  - This is a non-invasive kind of attack, and is the subject of intense research.






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Typical Countermeasures to Prevent Hardware Attacks

a) Obfuscate data in registers and buses
   - Scramble, encrypt, etc.

b) Obfuscate the IC layout
   - Use 3D stacking, dummy circuitry, etc.

c) Add metal mesh on top of the circuit.
   - If the circuit is probed, it will cause a short and the stored data resets.

d) Countermeasures against side channel attacks.
   - Random noise generator, secret hiding, etc.

e) Physical unclonable function (PUF)
   - Can be used to design low-overhead security protocols.






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Hardware Trojan

- Malicious logic inserted into a circuit without the knowledge of the designer / user.
  - Carries a trigger condition and a payload (may be malicious).
  - Very difficult to detect.
- Trojans can also be used for defensive purposes.
  - Any unauthorized change in the circuit will be detected.
  - Can be used for copyright protection (IC fingerprinting).






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To Summarize …

- A hardware implementation of a security device may be based on well-known secure algorithms.
  - The implementation of the hardware may be faulty, resulting in vulnerabilities.
  - The attacker tries to exploit the vulnerabilities.
- Next generation security chips will include countermeasures to protect against such attacks.

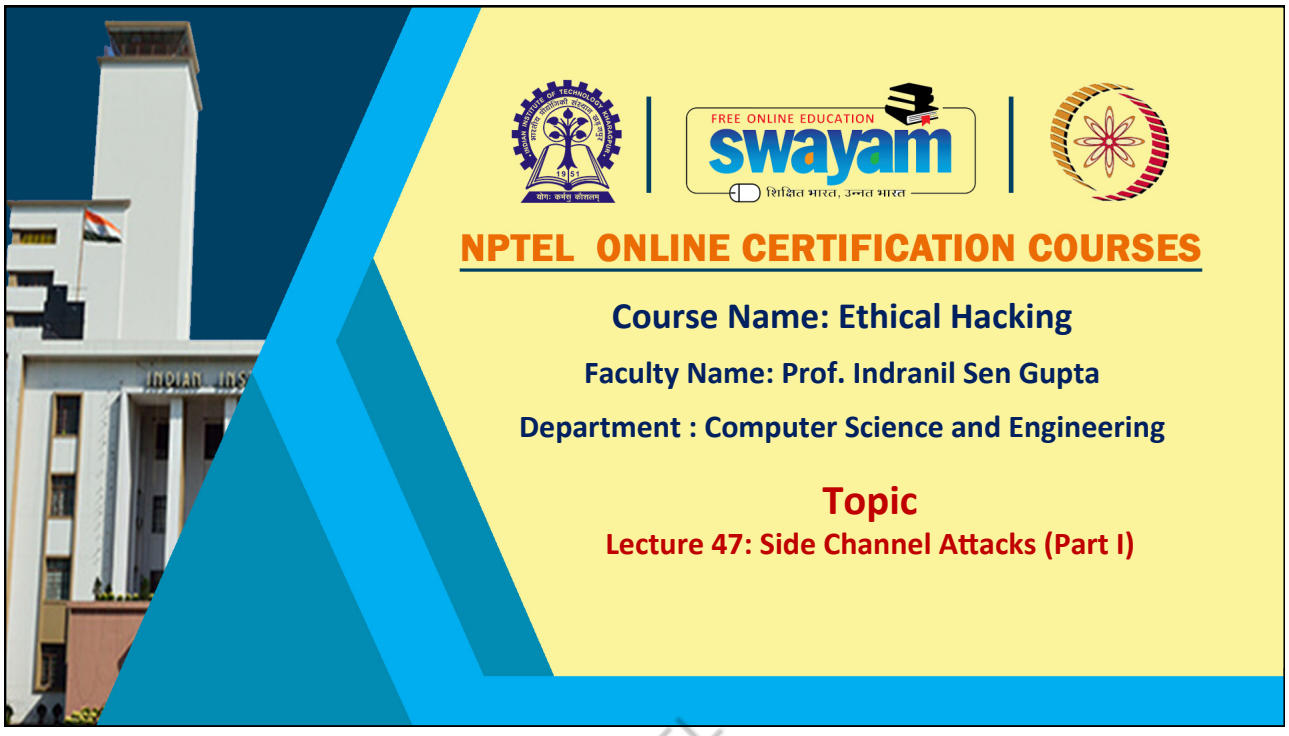





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Faculty Name: Prof. Indranil Sen Gupta

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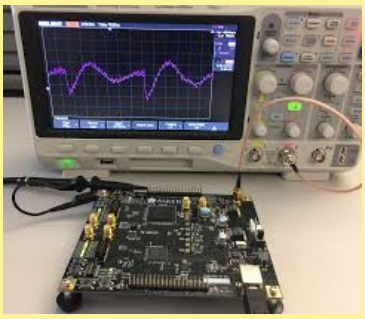
Lecture 47: Side Channel Attacks (Part I)

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CONCEPTS COVERED

- Side channel attack
- Timing analysis attack

Side Channel Attacks



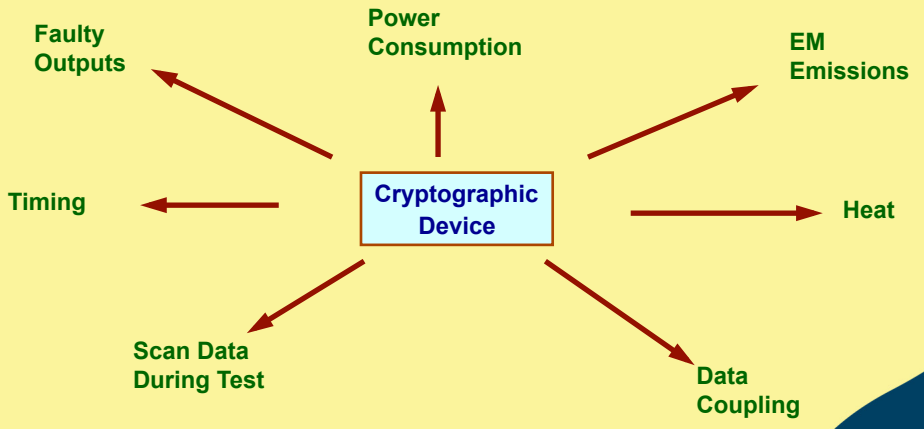
- Side channel attack / cryptanalysis:
  - New research area of applied cryptography.
  - Gained momentum since mid nineties.
  - Basic idea:
    - ❖ Capture unintended leakage of information during operation.
    - ❖ Can be exploited to extract key with relatively low effort.

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Why Important?

- A developer of a secure product has to defend it against all possible attack paths.
- In side channel attack:
  - The mathematical security of the cryptographic algorithms is not being questioned.
  - It is the implementation of these algorithms that is at risk to be broken.

Typical Side Channels



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Timing Analysis Attack

Introduction

- Basic concept
  - The first side-channel based attack to be published [Paul Kocher, 1995].
  - Attacker tries to break a cryptosystem by analyzing the execution time for the overall cryptographic operation.
- What does it try to exploit?
  - Computation time for a private key operation is dependent on the key in some way.
  - Particularly true for asymmetric key algorithms.






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An Example

- Square-and-multiply algorithm for modular exponentiation (used in RSA, Diffie-Hellman).
  - Execution time depends linearly on the number of '1' bits of the key.
  - Repeated executions with the same key and different inputs can be used.
    - ❖ To perform statistical correlation analysis of timing information.
    - ❖ The key can be recovered completely.






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- **Square-and-add exponentiation.**
  - How to calculate x^n ?

```
Power (x,n)  =  x,                          if n = 1
             =  Power (x^2, n/2),           if n is even
             =  x . Power (x^2, (n-1)/2),   if n>2 is odd
```

  - **Advantage**:
    - Simple implementation requires *(n-1)* multiplications.
    - This algorithm uses only $O(\log_2 n)$ multiplications.






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- Illustration:

$$x^{13} = x^{1101}$$
$$= x^{(1*2^3 + 1*2^2 + 0*2^1 + 1*2^0)}$$
$$= x^{2^3} * x^{2^2} * 1 * x^{2^0}$$
$$= x^8 * x^4 * x^1$$
$$= (x^4)^2 * (x^2)^2 * x$$
$$= (x^4 * x^2)^2 * x$$
$$= ((x^2)^2 * x^2)^2 * x$$
$$= ((x^2 * x)^2)^2 * x$$

- Requires only 3 squarings and 2 multiplications rather than 12 multiplications.
  - Number of squarings and multiplications can directly give the ***number of 1's*** in the key.





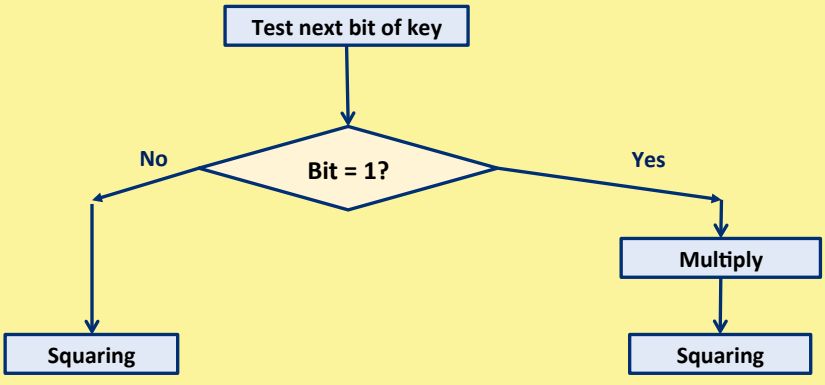

- Pseudo-code to compute $b^e \pmod m$

```
Bignum modpow  (Bignum b, Bignum e, Bignum m)  {
   Bignum result = 1;
   while (e > 0) {
     if (e & 1 > 0)  result = (result * b) % m;
     e = e >> 1;
     b = (b * b) % m;
   }
   return result;
}
```



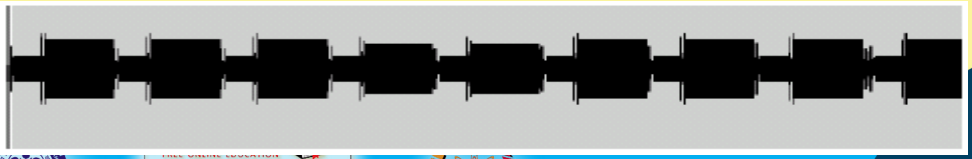
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- Modified algorithm -- make branches symmetric:

```
Bignum modpow  (Bignum b, Bignum e, Bignum m) {
  Bignum result = 1;
  while (e > 0) {
    if (e & 1 > 0)     result = (result * b) % m;
    else  a = (b * c) % m;
    e = e >> 1;
    b = (b * b) % m;
  }
  return result;
}
```



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- Timing analysis can reveal the number of 1's in the secret key.

  Time = n * t_{square} + k * t_{mul}

  n : number of bits in the key

  k: number of one bits in the key

  t_{square} : time to compute square

  t_{mul} : time to compute multiplication

- The suggested countermeasure can make the time independent of the key.

  Time = n * (t_{square} + t_{mul})

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What it means?

- If the device carrying out the cryptographic operation is available for analysis …
  - We can gain valuable insight into the internal execution.
  - For RSA, the complexity of brute-force attack can be drastically reduced.
- Security implications:
  - We use various sorts of smart cards in our daily life.
  - Side-channel attack can pose a serious threat.
  - Secure side-channel attack resistant implementations are necessary.






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- An algorithm may be mathematically very secure.
- But weaknesses in hardware or software implementations may make it vulnerable against side-channel attacks.
  - Secure implementation is very important in this context.

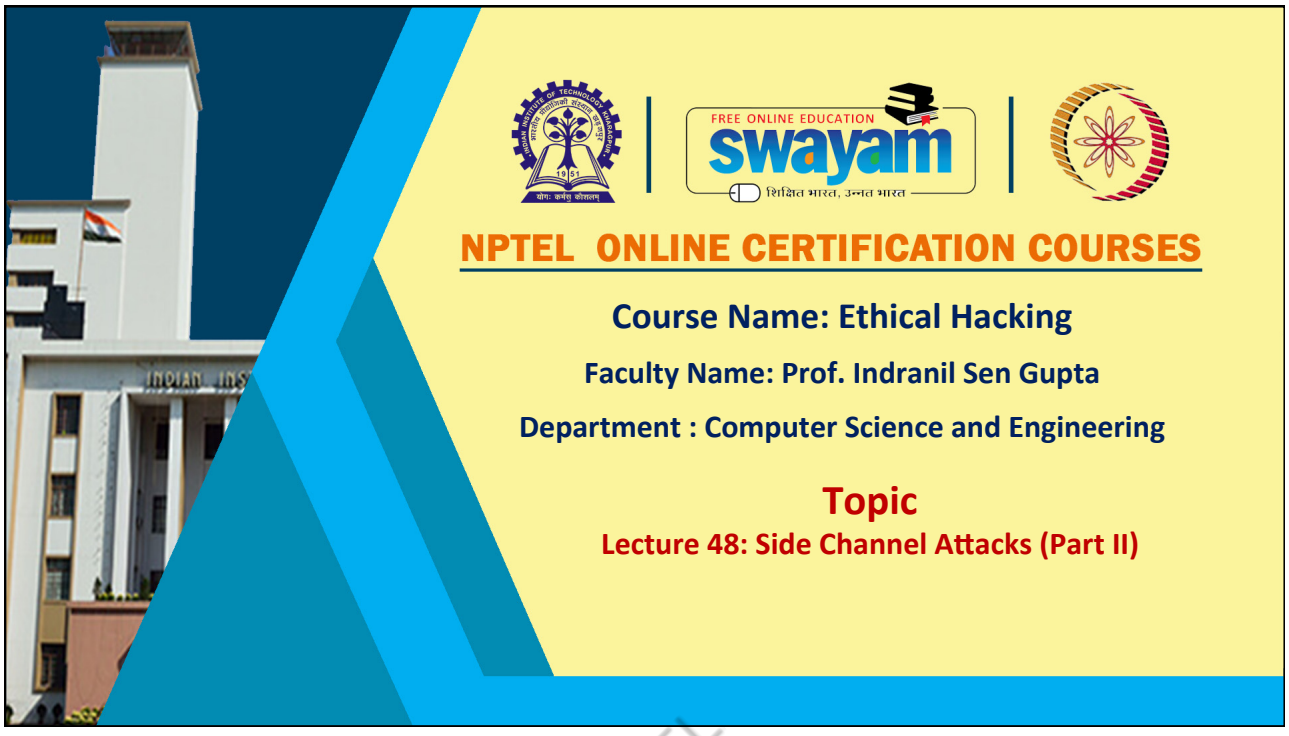





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CONCEPTS COVERED

- Power analysis attack
- Simple and differential power analysis
- Countermeasures

Power Analysis Attack






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Introduction

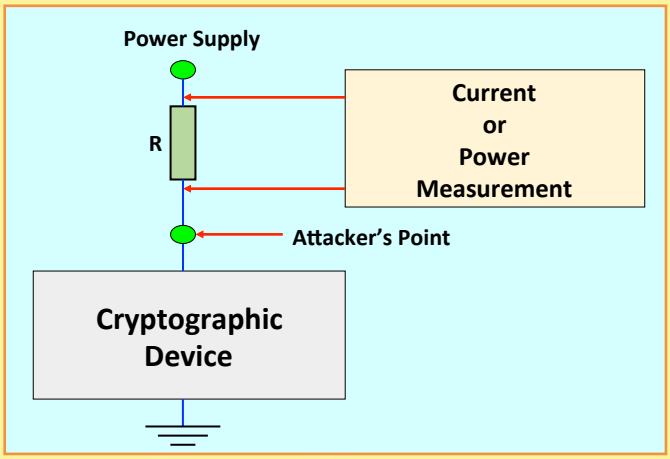
- Basic concept:
  - A much more effective form of side channel attack [Paul Kocher et al, 1998].
  - Analyzes the power consumed by a device during the processing of some cryptographic operation.
- What it can yield?
  - Information about what the device is doing.
    - ❖ Can extract the key information.



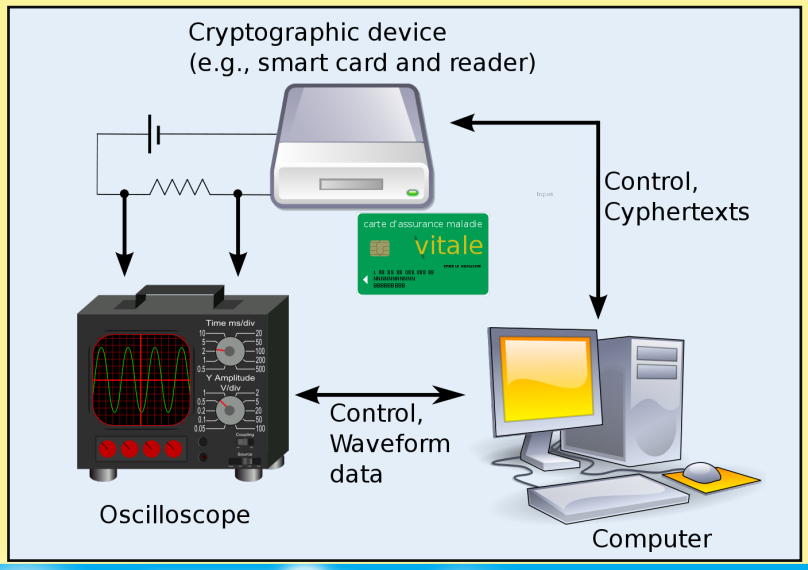



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Data Acquisition Setup



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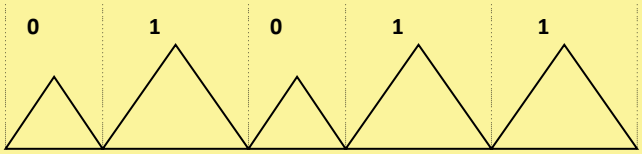
Simple Power Analysis (SPA)

- Attacker directly uses power consumption to learn bits of secret key.
  - Waveforms visually examined.
- Can identify:
  - Big features like rounds of DES/AES, square vs. multiply in RSA exponentiation.
  - Small features, like bit value.
- Relatively easy to defend against.

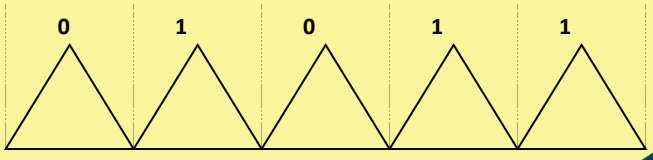
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How SPA Works?

Key = 101011
Square-and-Multiply Algorithm



With "Dummy" Operations



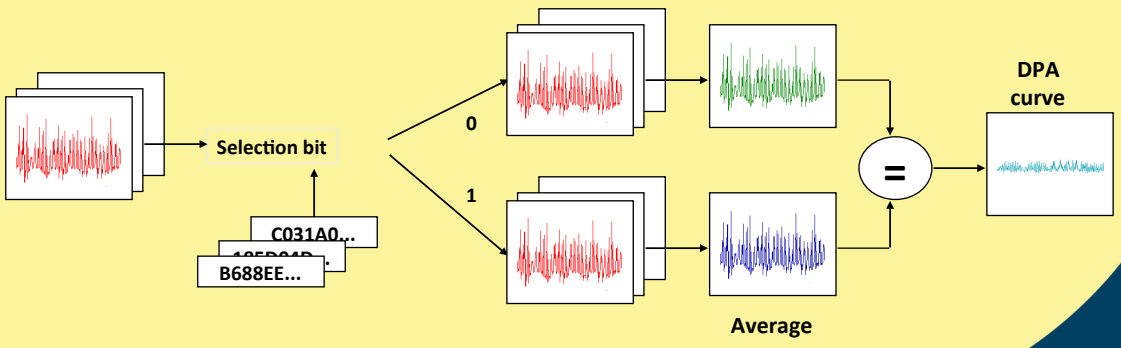
Power Traces

Differential Power Analysis

- More complex.
- Partition the data and related curves into two sets according to selected bits.
- Take the difference, and look for peaks or differences.

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DPA Process Schematic



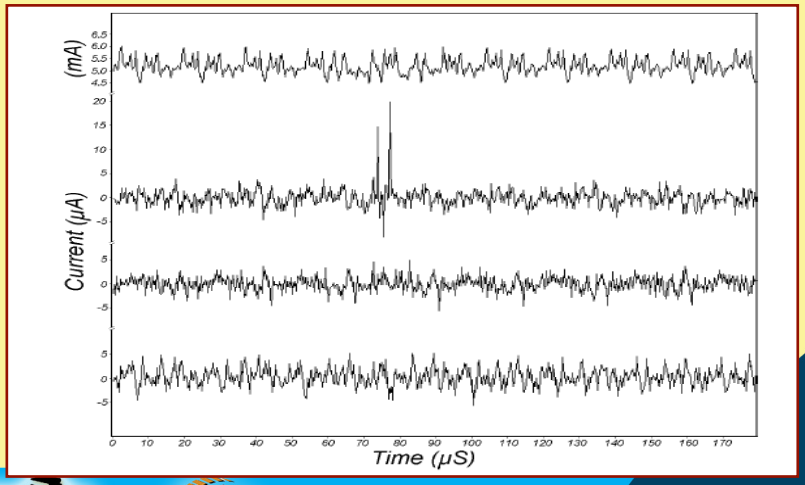
DPA Result Example

Average Power Consumption

Power Consumption Differential Curve With *Correct* Key Guess

Power Consumption Differential Curve With *Incorrect* Key Guess

Power Consumption Differential Curve With *Incorrect* Key Guess



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Countermeasures

Introduction

- Relatively easy to implement for timing analysis.
  - Make the execution time data independent.
- Power analysis attacks that look at specific intermediate values of the implementation are much harder to defeat.
  - Two broad approaches practiced:
    - **a) Hardware-based**
    - **b) Software-based**





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a) Hardware-based countermeasures:

- Special logic styles that minimizes data-dependent leakage.
- Implementation of masking schemes.
- Addition of noise with noise-generators.
- Random process interrupts that provide for an internal timing de-synchronization.





b) Software-based countermeasures:

- Aim to avoid the occurrence of predictable intermediate results.
- Introduce redundant computations.
- Internal randomization used to mask the data representation used.
  - ❖ A random value, not known to the attacker, is added or multiplied with intermediate values.

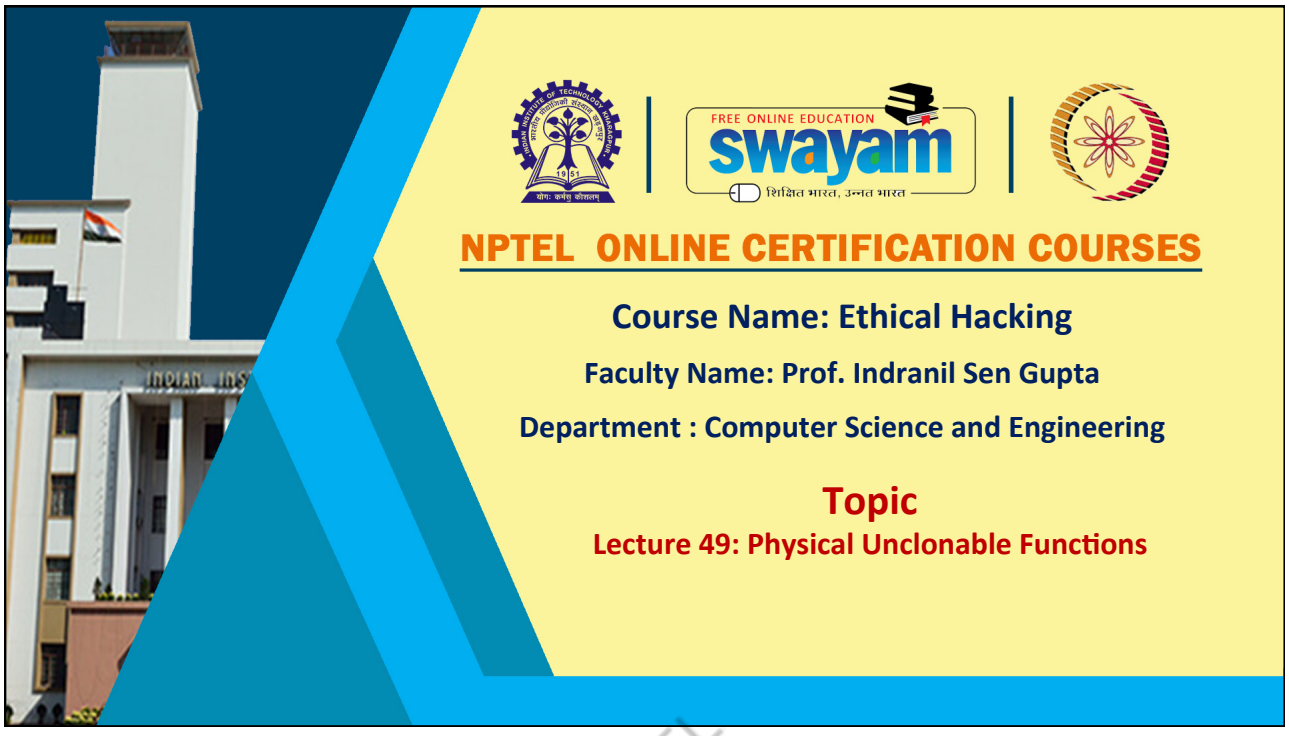
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Conclusion

- Side channel attacks
  - Powerful technique but specific.
    - ❖ Targets a particular implementation rather than a generic algorithm.
    - ❖ Most devices as well as software implementations on embedded platforms can be targeted.
    - ❖ Hard to evaluate and prevent.
  - Possible loophole:
    - ❖ Resisting one kind of attack may introduce weaknesses with respect to another one.
  - An active area of research.


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Course Name: Ethical Hacking

Faculty Name: Prof. Indranil Sen Gupta

Department : Computer Science and Engineering

Topic

Lecture 49: Physical Unclonable Functions

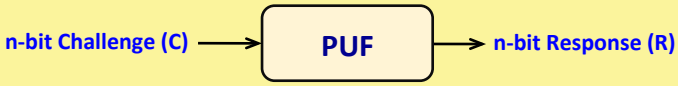
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CONCEPTS COVERED

- Physical Unclonable Function (PUF)
- Applications of PUF
- PUF implementation

What is a PUF?

- Fingerprint of some device.
  - A challenge-response mechanism in which the mapping between an applied input ("*challenge*") and the corresponding observed output ("*response*") is dependent on the complex and variable nature of a physical material.
  - The challenge-response mapping is *unclonable* (ideally) and *instance-specific* (depends on manufacturing process variations evident in ASICs).



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Some Desirable Properties

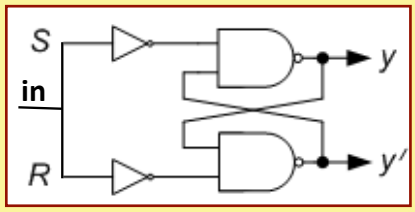
- **Evaluatable**:
  - Given *PUF* and *x*, it is easy to evaluate *y = PUF(x)*.
- **Unique**:
  - The *PUF(x)* contains some information about the identity of the physical entity embedding the *PUF*.
- **Unclonable**:
  - Given *PUF*, it is hard to construct a procedure *PUF'*, where *PUF ≠ PUF',* and *PUF'(x) = PUF(x)* for all *x*.
- **One-way**:
  - Given only *y* and the corresponding *PUF* instance, it is hard to find *x* such that *PUF(x) = y.*

4

An Example with a Simple S-R Latch

- Make the input *in=1*.
  - We shall get *y=1, y'=1*.
- Now make the input *in=0*, both of the following states are possible:
  - *a) y=1, y'=0*
  - *b) y=0, y'=1*

Source of randomness



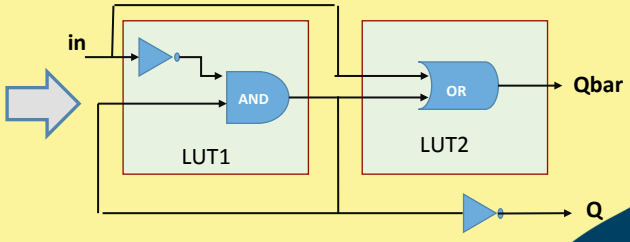
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From Theory to Practice

- FPGAs are ideal for security implementations.
  - In-house and high-performance.
  - Programmability is an added feature.
  - But careful implementation is needed.

The non-determinism and hence the randomness is gone!

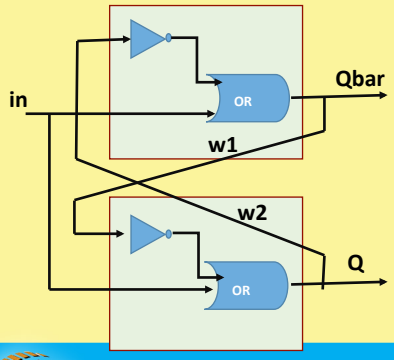
```
module SR (in, Q, Qbar);
  input in;
  output Q, Qbar;
  nand N1(Q, ~in, Qbar);
  nand N2(Qbar, ~in, Q);
endmodule
```



Another Attempt

- This design has the non-determinism as expected!
- We can also design using NAND primitives.

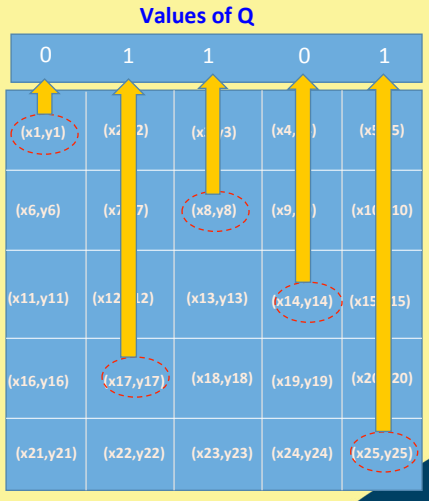
```
module SR (in, Q, Qbar);
  input in;
  output Q, Qbar;
  wire w1, w2;
  nand N1(Q, ~in, w1);
  nand N2(Qbar, ~in, w2);
  assign w1 = Qbar;
  assign w2 = Q;
endmodule
```



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The Silicon Space

- Mismatch in driving capabilities of the gates.
- Difference in routing delays of the feedback path.
  - A latch cell will give either 0 or 1 as output.
  - Depends on the (x,y) position of the silicon area.



Advantages of PUF

PUF as Security Primitive

- **Reduce Cost**
- **Increase Security**

World without PUF

- Trusted party embeds and tests secret keys in a secure non-volatile memory (NVM).
- EEPROM adds additional complexity to manufacturing.
- Adversaries may physically extract secret key from non-volatile memory.

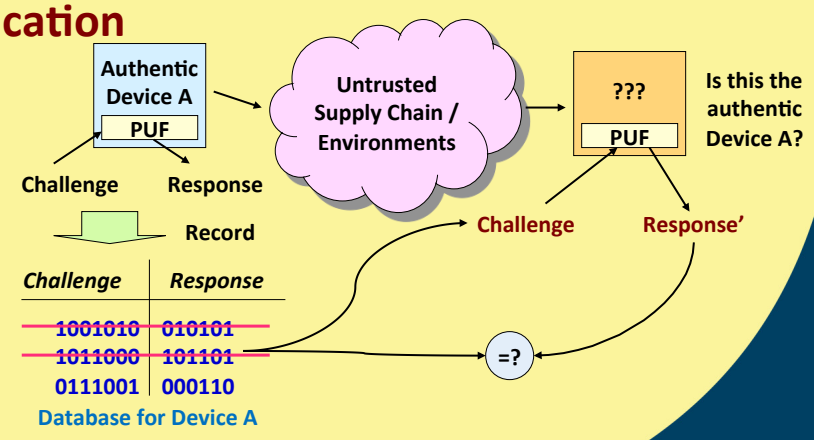
World with PUF

- Intrinsic properties of device is used to generate secret key.
- Key never leaves the IC's cryptographic boundary, nor be stored in a non-volatile memory.
- Key is deleted after usage in encryption or decryption process.

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PUFs for Identification

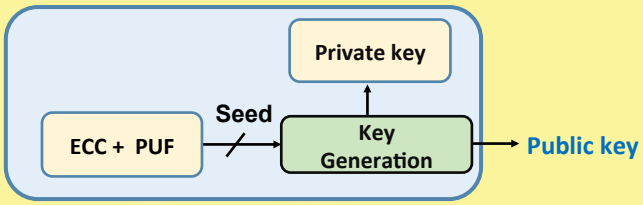
- Protects against ASIC substitution and counterfeits without using cryptographic operations.



IC rejected if *Response'* does not match the enrolled *Response*

PUFs for Private/Public Key Pair Generation

- PUF response is used as random seed to a private/ public key generation algorithm.
- No secret needs to be handled by a manufacturer.
- A device generates a key pair on-chip, and outputs a public key.
- The public key can be endorsed at any time.



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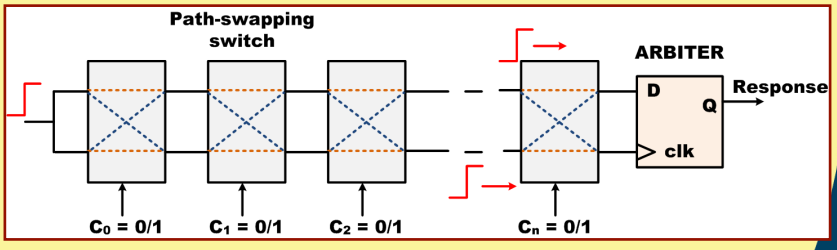
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Practical Designs of PUF

- We are interested in Silicon-PUF circuits
  - Utilize the unavoidable and inpredictable process variation effects of modern deep-submicron MOSFET devices.
  - From circuit design perspective, process variation is a challenge, but is useful for PUF design.
- Various designs of PUFs have been explored.

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(a) Arbiter PUF

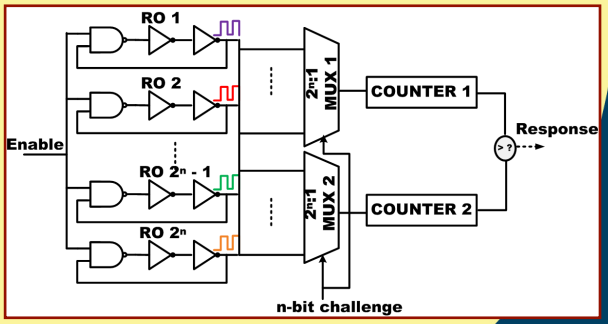


- Composed of *n* two-port switching stages, for an *n*-bit challenge size.
  - Number of possible paths is 2^n.
  - A challenge selects a unique path.
  - Accumulated delay at the end of the path is compared by an arbiter circuit, which gives a 1-bit decision.

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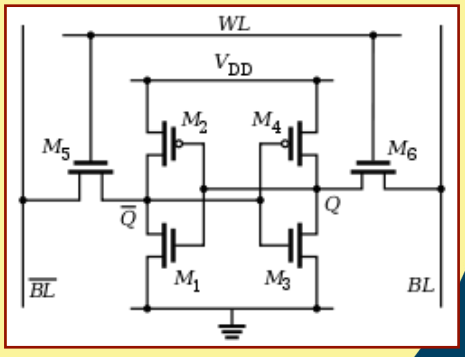
(b) Ring Oscillator PUF (ROPUF)

- An n-bit challenge selects two different ROs from a bank of 2^n ROs.
- Process variation results in ROs to have different oscillation frequencies.
- Compare frequencies of two oscillators using counters.
- A comparator generates the final decision.



(c) SRAM PUF

- Power-up initial value of SRAM cell can be used as response; cell address is the challenge.
- SRAM fabrication compatible with digital logic process in regular ICs.
- FPGA implementation of SRAM PUF is very difficult.
  - Since SRAM modules are cleared by default on power-up.



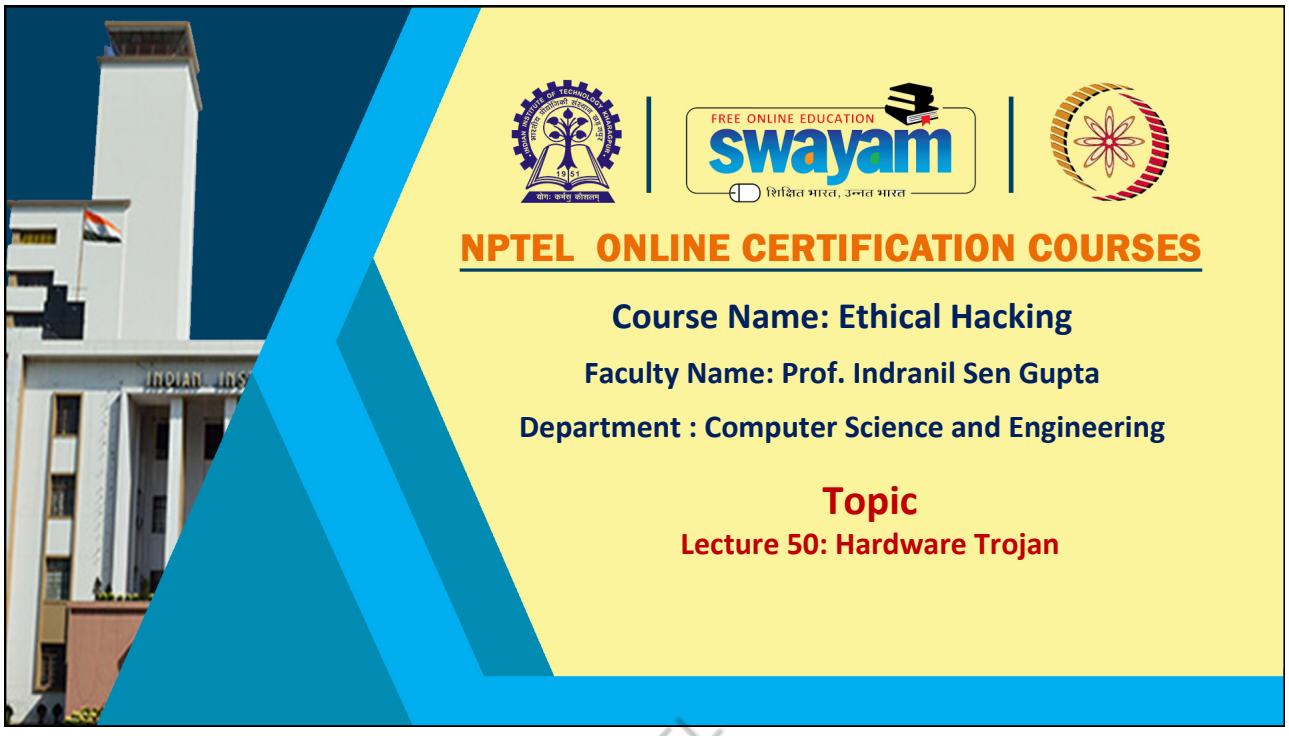
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Summary

- PUFs are not very expensive to realize.
- Many recent security protocols are based on PUFs.
  - Makes it difficult to mount hardware-based attacks.


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Course Name: Ethical Hacking

Faculty Name: Prof. Indranil Sen Gupta

Department : Computer Science and Engineering

Topic

Lecture 50: Hardware Trojan

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CONCEPTS COVERED

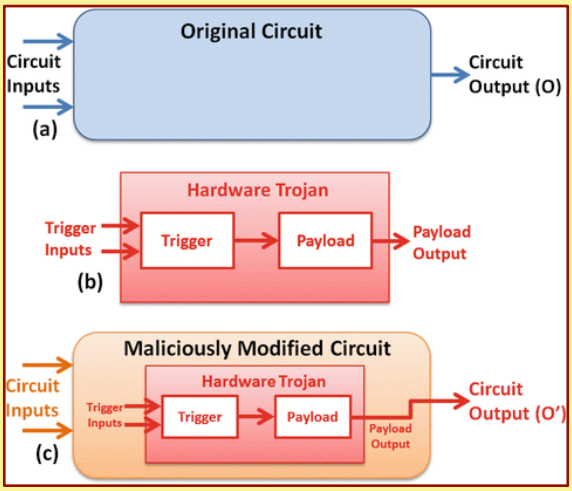
- What is hardware trojan?
- Types of trojans
- Trojan detection

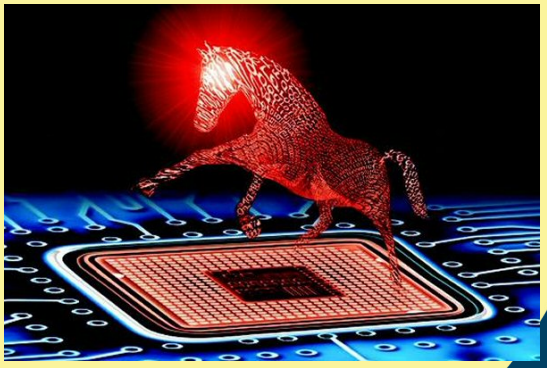
What is Hardware Trojan (HT)?

- It is a malicious modification of the circuitry of an IC chip.
  - During design of fabrication.
- A HT is completely characterized by its physical representation and its behavior.
- The payload of a HT is the entire activity that the trojan executes when it is triggered.



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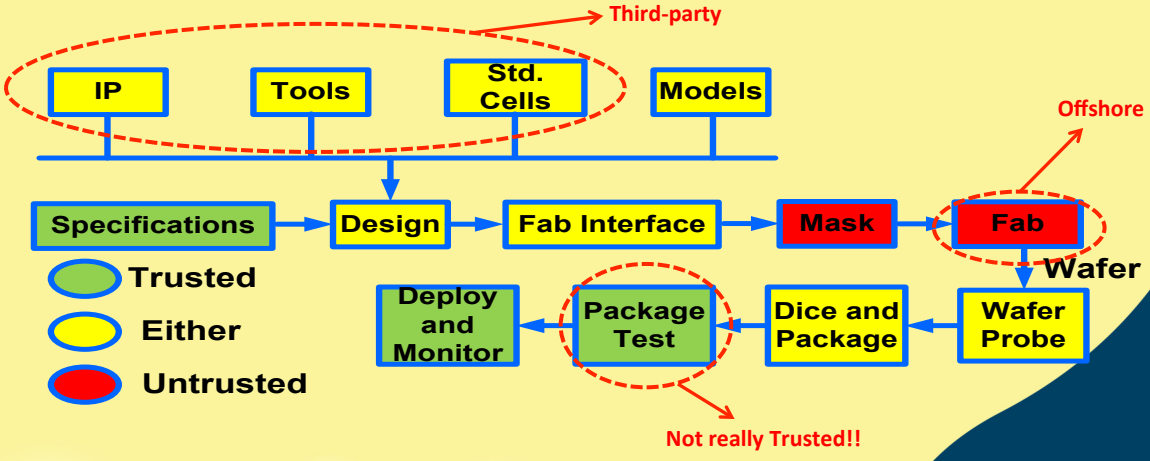


Effects of Prevalent Practices

- Prevalence of IP based design.
- Routine use of CAD tools from EDA vendors.
- Fabless manufacturing model (trend on the rise).
- Outsourcing of manufacturing to offshore fabs.
- Loss of control over design and manufacture.
- Potentially untrusted parties getting involved.

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Modern IC Design and Manufacturing



Hardware Trojans really are …

- **Malicious modifications to design**
  - Can take place pre or post manufacturing.
  - Inserted by intelligent adversary.
  - Extremely small hardware overhead.
  - Stealthy => difficult to detect.
  - Causes IC to malfunction in-field.
- **Results**:
  - Potentially disastrous consequences.
  - Loss of human life and property.






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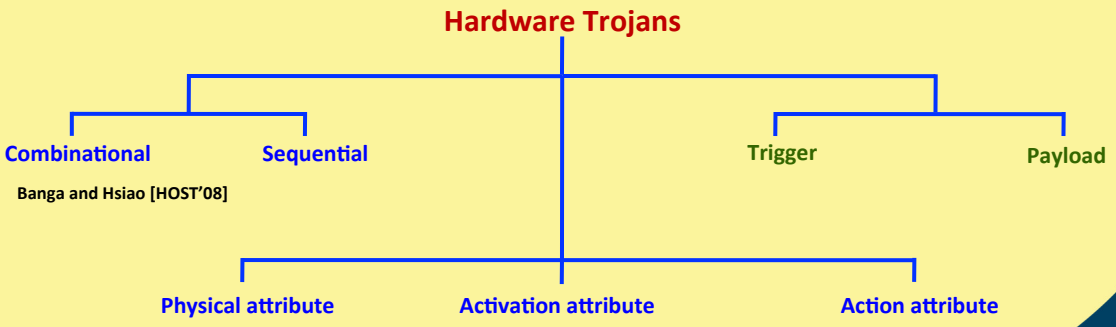
How Realistic are Hardware Trojans?

- Do hardware trojans really exist?
  - No concrete proof obtained as yet.
  - Tampering masks in fab is not easy (highly complex).
  - Reverse-engineering a single IC can take months.
- But there is strong evidence they do….
  - Numerous suspected military / commercial cases (as early as 1976!!).
  - Reverse-engineering of ICs is widely believed to be performed by reputed companies (IBM has patents).
  - Highly sophisticated commercial software tools for reverse-engineering are available (Chipworks, etc.), and academic efforts (Cambridge University).





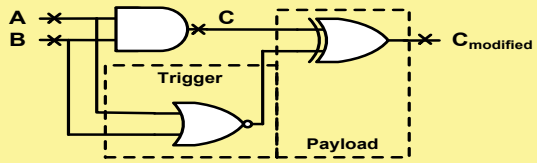

Trojan Taxonomy



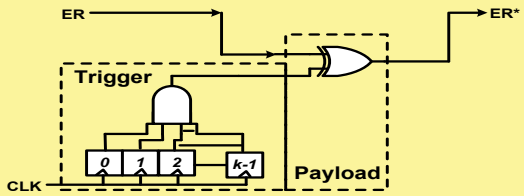
Banga and Hsiao [HOST'08]

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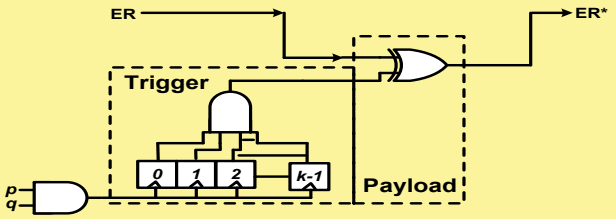
Digital Trojans



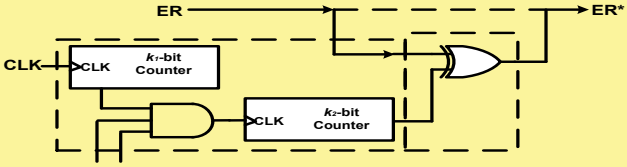
Combinational Trojan (simplest)



Sequential (Synchronous) Trojan ("Time Bomb")



Sequential (Asynchronous) Trojan



Hybrid Trojans

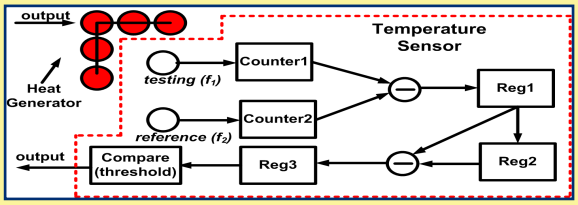




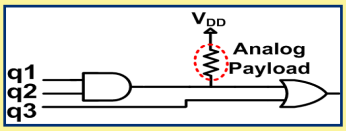
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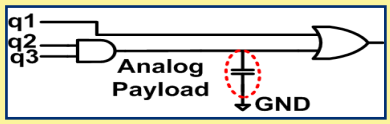
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Analog Trojans



Analog Trojan (activity-triggered)





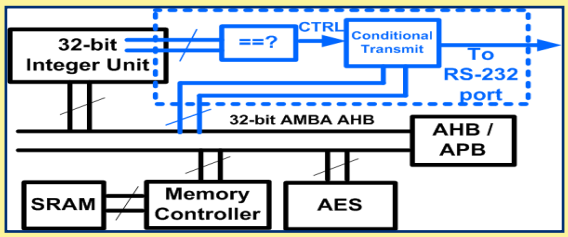
Analog payload Trojan



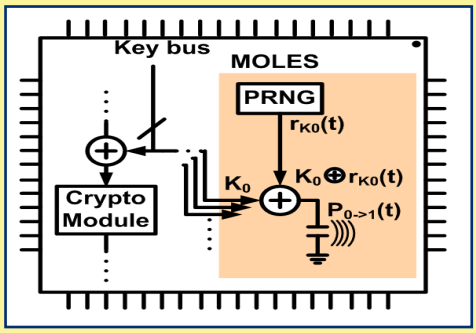


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Information Leakage Trojans



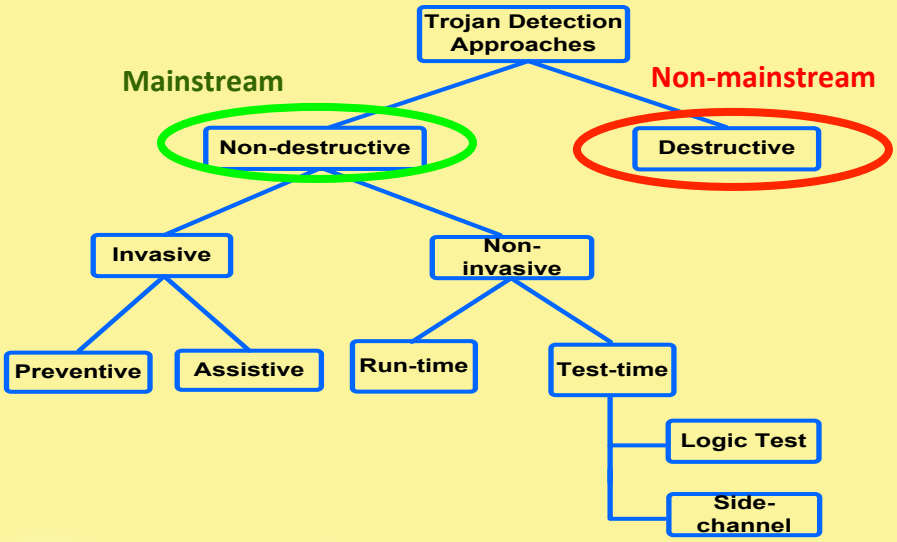
Logic-value Based



Side-channel Leakage Based

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Approaches to Trojan Detection



Why is Trojan detection difficult?

- High overhead.
- Trojans are stealthy.
- Trigger nodes have very low observability.
- Extremely large number of Trojan types possible.
- No single method can detect all types of Trojans.






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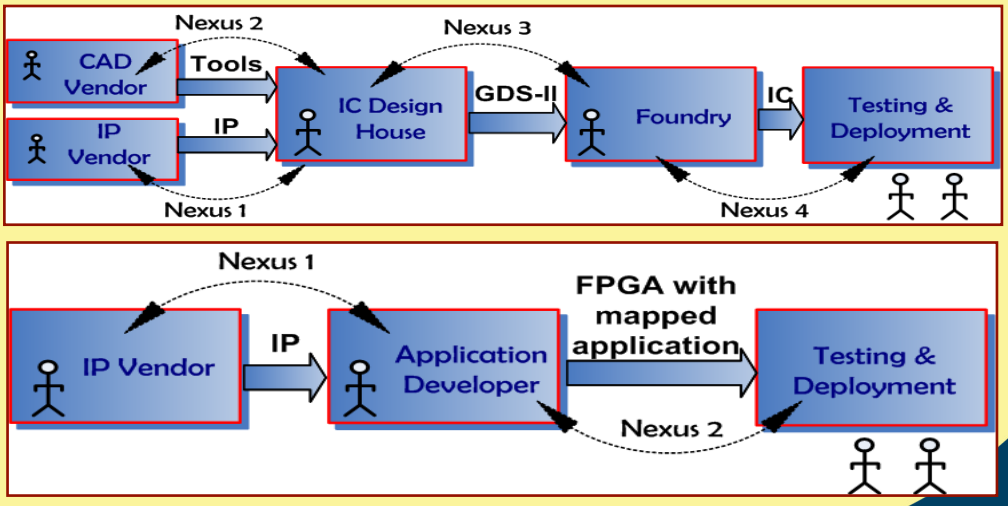
Multi-Level Attack

- Uses nexus between multiple parties.
- Only parties which are part of the nexus can benefit.
- The nexus eases the burden on individual parties.
- Additional challenges to detect.






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Conclusion

- IC design/manufacturing practices are insecure.
  - Third-party IPs and off-shore manufacturing.
  - Potentially *untrusted* parties pay a major role.
- Hardware Trojans are malicious circuit modifications.
  - Small overhead, hugely destructive impact.
  - Difficult to detect by traditional testing means.
- State-of-the-art:
  - Both design and test techniques have been proposed.
  - Effectiveness of the proposed techniques limited to the particular types of Trojans.


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